

Gradient Descent

- ❑ A commonly used optimization function that adjusts weights according to the error they caused.
- ❑ Gradient is another word for slope, and slope, in its typical form on an x-y graph, represents how two variables relate to each other.
- ❑ In NNs, the slope we care about describes the relationship between the network's error and a single weight: how does the error vary as the weight is adjusted?
 - ❑ Which weight will produce the least error?
 - ❑ Which one correctly represents the signals contained in the input data, and translates them to a correct classification?
 - ❑ which one can see “eyes and nose” in an input image, and know that should be labeled as a face and not a dish?

Gradient Descent (cont.)

- ❑ As a neural network learns, it slowly adjusts many weights so that they can map signal to meaning correctly.
- ❑ The relationship between network **Error** and each of those **weights** is a derivative, dE/dw , that measures the degree to which a slight change in a weight causes a slight change in the error.
- ❑ Each weight is just one factor in a deep network that involves many transforms.
- ❑ The signal of the weight passes through activations and sums over several layers, so we use the chain rule of calculus to march back through the networks activations and outputs and finally arrive at the weight in question, and its relationship to overall error.

Gradient Descent (cont.)

- The chain rule in calculus: $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$
- In a feedforward network, the relationship between the net's error and a single weight can be presented as:

$$\frac{dError}{dWeight} = \frac{dError}{dActivation} \cdot \frac{dActivation}{dWeight}$$

- Given two variables, **Error** and **Weight**, that are mediated by a third variable, **Activation**, through which the weight is passed, you can calculate how a change in Weight affects a change in Error by first calculating how a change in Activation affect a change in Error, and how a change in Weight affects a change in Activation.